

Real_LTSpice_Netlist_2

Prompt

What should i plot in LTSpice to verify if the simulation works correctly ?

LTSpice AM Radio: What to Plot and Verify

To verify that the AM radio netlist is working correctly, you need to follow the signal through the circuit from the antenna to the headphones. In LTSpice, run the simulation (`.tran 0 5m 0 100n`), right-click the black waveform window, and click **Add Traces**.

Here are the 4 critical nodes to plot and exactly what you should expect to see at each stage:

1. The Input Signal (Antenna/Tuning Stage)

- **What to plot:** `V(in)` or `V(rf_in)`
- **What you are verifying:** That the arbitrary waveform generator is successfully creating an AM radio signal.
- **What you should see:** A high-frequency signal (the 693 kHz carrier wave) whose height/amplitude continuously grows and shrinks. The "shape" of this growing and shrinking (the envelope) corresponds to the 1 kHz audio tone. It should be very small in amplitude (around 10mV to 20mV peak-to-peak).

2. The RF Amplifier Output

- **What to plot:** `V(rf_c)`
- **What you are verifying:** That the first transistor (Q1) is successfully amplifying the weak antenna signal without distorting it.
- **What you should see:** The exact same AM wave shape as `V(in)` , but much larger (amplified). Because it's a common-emitter amplifier, the signal will be riding on a DC DC voltage (usually somewhere between 3V and 6V, rather than centered at 0V).

3. The Envelope Detector Output

- **What to plot:** `V(det_out)`
- **What you are verifying:** That the diode (D1) and the RC filter (`R_det` and `C_det`) are stripping away the high-frequency carrier wave and leaving only the audio.
- **What you should see:** The high-frequency "blur" should be gone. You should now see a distinct, smooth 1 kHz sine wave. It will still be riding on a small DC offset voltage. If the

wave looks jagged or like a sawtooth, your RC time constant is wrong (but the 10nF/10kΩ combo in the netlist should make it look smooth).

4. The Final Audio Output (Headphones)

- **What to plot:** $V(\text{out})$
- **What you are verifying:** That the second transistor (Q2) amplifies the weak audio signal and that the final coupling capacitor (C2) successfully blocks the DC voltage.
- **What you should see:** A strong, clean 1 kHz sine wave perfectly centered exactly at 0V. The amplitude should be significantly higher than $V(\text{det_out})$. This is the final audio signal that drives the 32-ohm speaker load.

Quick Troubleshooting Tip:

If you plot $V(\text{out})$ and it is just a flat line or heavily clipped (squared off at the top or bottom), check the DC bias of $V(\text{af_c})$. If $V(\text{af_c})$ is sitting at 9V or near 0V, the transistor is saturated/cut-off, and you would need to adjust the 10k base bias resistor ($R_{\text{af_b}}$). The provided netlist values should keep it safely in the active region.

[Real_LTSpice_Netlist_3](#)